

```
In [4]: import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('maxent_ne_chunker')
nltk.download('maxent_ne_chunker_tab')
nltk.download('words')

# Sample news headlines
headlines = [
    "Prime Minister Narendra Modi visited USA",
    "Google launches new AI model",
    "Virat Kohli scored a century in Mumbai"
]

for headline in headlines:
    print("\nHeadline:", headline)

    # Tokenize
    words = nltk.word_tokenize(headline)

    # POS tagging
    pos_tags = nltk.pos_tag(words)

    # Named Entity Recognition
    ner_tree = nltk.ne_chunk(pos_tags)

    print("Named Entities:")
    print(ner_tree)
```

```
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data] C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package maxent_ne_chunker to
[nltk_data] C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data] Package maxent_ne_chunker is already up-to-date!
[nltk_data] Downloading package maxent_ne_chunker_tab to
[nltk_data] C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data] Unzipping chunkers\maxent_ne_chunker_tab.zip.
[nltk_data] Downloading package words to
[nltk_data] C:\Users\Priyansh\AppData\Roaming\nltk_data...
[nltk_data] Package words is already up-to-date!
```

Headline: Prime Minister Narendra Modi visited USA

Named Entities:

```
(S
  Prime/NNP
  Minister/NNP
  (PERSON Narendra/NNP Modi/NNP)
  visited/VBD
  (ORGANIZATION USA/NNP))
```

Headline: Google launches new AI model

Named Entities:

```
(S (GPE Google/NNP) launches/VBZ new/JJ AI/NNP model/NN)
```

Headline: Virat Kohli scored a century in Mumbai

Named Entities:

```
(S
  (PERSON Virat/NNP)
  (PERSON Kohli/NNP)
  scored/VBD
  a/DT
  century/NN
  in/IN
  (GPE Mumbai/NNP))
```

In []: